

The project

What we expect, and how a
proposal is approved

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ADMIN

The whole grade, on one slide

The project	one, individual, 100 % of the grade
Proposal	200 words, by Monday 9 November ; work starts after sign-off
Submission	a GitHub repository: paper, code, data, 15-minute recording
Deadline	Sunday 20 December, 23:59 . Late work is not accepted.

Both documents below are on unil-ada.github.io/project.html

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One set of rules, one set of advice

Requirements

The rules. Deadlines, what you submit, how it is assessed, AI tools. This is the only binding document.

"Starting and shipping your project"

Our advice. How we would go about it: choosing a question, finding data, organising the repository. Its title says "guidelines", but it is advice, not a checklist.

Following the advice does not add up to a grade, and it is not a template to fill in. If the two ever seem to disagree, the Requirements win.

What a project is

One question you care about, data you actually have, methods from the course applied with judgement, and a paper that lets a stranger reproduce what you did.

The grade rewards evidence that **you** can apply what the course taught. Not the size of the data, not the number of models, not a positive result.

Negative results are fine. A hypothesis tested honestly that finds nothing is not marked down for the result.

We have already received proposals

Thank you: that is keen, and it is early.

Some of them read like our Guidelines, rewritten: the right three paragraphs, the right phrases, methods we will not teach until late November.

A proposal that reproduces our Guidelines tells us that you can read our Guidelines. It does not tell us anything about you, your question, or your data.

So this year, every proposal comes with five minutes

1. You email the proposal, as the Requirements say: 200 words, question, data, method.
2. **You talk to me for five minutes.** Every student, in person, with your data open in Nuvolos.
3. Sign-off comes by email **after** the conversation.

No conversation, no sign-off. The deadline for the proposal stays **9 November**.

What the five minutes are

- **Show me your data.** How many rows, what is missing, what surprised you.
- **Your question, in your words.** What result would count as an answer, either way?
- **Why this method,** and what you will do if it does not work.

It is not an exam. You are not expected to know methods we have not taught yet, and it is not graded.

You are expected to know your own question and your own data. Nobody can do that for you, and nothing can.

When

- in the **exercise sessions** in October: project time, first come, first served;
- in **office hours**, Fridays from 13:00, by appointment: email to book.

Already sent a proposal? You are first in line. Book five minutes, bring your data.

Not yet? Do not write the proposal first. Get the data, look at it, then write 200 words about what you saw.

AI tools: allowed, expected, declared

- Use them for code and for writing. Declare every tool, and what you used it for, in the appendix of your paper.
- **Undeclared use is misconduct.** Declared use is normal practice.
- **You answer for everything you submit.** The recording, like the five minutes, is you explaining your decisions. If you cannot explain a choice, it was not yours.

What a chatbot cannot give you: **a question you actually care about, and data you have looked at yourself.**

Kaggle, UCI, built-in datasets

- **Fine as a source** of raw data, and fine for a first model.
- **Weak as the whole project**: they arrive clean, everyone has seen them, and the analysis already exists online.

Better: data that needs work. FRED, World Bank, OECD, Eurostat, ECB, **SNB**, the **Swiss Federal Statistical Office**; daily prices through `yfinance`; filings on SEC EDGAR. Two sources merged, a variable you had to construct, missing values you had to decide about. That work is graded.

What makes a question yours

- a question you were already curious about, brought to data; or
- a known question asked of data nobody asked it of: a different country, market, period or sample.

It does not need to be new to the literature. It does need decisions in it that are yours, and that you can explain.

Show us what the course gave you

The project is where you show what you learned here. Regression is where the course starts, not where it ends: over the semester you will also meet classification, neural networks, recurrent models, Gaussian processes, dimensionality reduction, clustering and reinforcement learning.

The projects we enjoy most bring in, from that range, what fits their question.

- Choose methods **because they fit your question**, not to fill a list.
- Start simple, then let the simple model show you what it cannot do.
- A method you can explain beats three you cannot. Everything must run in your Nuvolos workspace.

When the methods arrive

Weeks	Methods
4–6 (5–25 Oct)	regression, classification
7–8 (26 Oct – 8 Nov)	neural networks, recurrent models
10–12 (16–30 Nov)	Gaussian processes, PCA, clustering

Start with what you can fit in October. A later method can be added once you have learnt it, and once the simple model has told you what it cannot do.

Last ten minutes: a first look at data

In Nuvolos: **File → New → Notebook**, then type this. Replace the link with your own dataset if you have one.

```
import pandas as pd

url = "https://fred.stlouisfed.org/graph/fredgraph.csv?id=DEXSZUS"    # CHF per USD, daily
df = pd.read_csv(url, parse_dates=[0], index_col=0)

print(df.shape)                # how many rows and columns?
print(df.isna().sum())         # how many gaps, and why?
df.plot()                      # one picture
```

Then write one sentence each: **what is one row, what is missing, what surprised you.**

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This week

1. Put your **full name** on your GitHub account.
2. Start a note called "**data I could use**". Every dataset you meet goes in it, with its URL.
3. Do the same first look for **one dataset of your own**: rows, gaps, one plot, three sentences.
4. That is what we will talk about in your five minutes.

Questions and bookings: anna.smirnova@unil.ch · Everything: unil-ada.github.io/project.html